Motor Starter

Category: **Electrical & Electronic Components & Kits** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: DOL STARTER

What it is used for

A direct-on-line (DOL) motor starter connects a suitable motor directly to the supply through switching and overload-protection components. Use it only with correctly rated equipment, proper wiring, and approved electrical isolation procedures.

Main parts

- Contactor and main switching contacts
- Thermal or electronic overload relay
- Start and stop push-buttons or control switch
- Enclosure and terminal blocks
- Control circuit and auxiliary contacts
- Supply terminals and motor output terminals

Before you start

1. Confirm the starter rating matches the motor, supply voltage, and application.
2. Inspect the enclosure, contactor, overload relay, terminals, and wiring for damage.
3. Isolate the electrical supply and apply lockout/tagout before opening the enclosure.
4. Verify absence of voltage using an appropriate tester and safe procedure.
5. Check motor earthing, cable sizing, overload settings, and protective-device coordination.
6. Confirm the motor and driven machine can start safely and the work area is clear.

How to use

1. Keep the supply isolated while inspecting or connecting the starter.
2. Check that all power and control terminals are correctly identified.
3. Connect the incoming supply, motor leads, and control circuit according to the approved wiring diagram.
4. Ensure protective earthing and required covers are correctly fitted.
5. Set the overload relay according to the motor nameplate and approved design.
6. Check that the stop control and emergency stop, where fitted, operate correctly.
7. Close the enclosure and remove tools before restoring power.
8. Restore supply only after authorised checks confirm the installation is safe.
9. Press the start control and observe the motor starting and running behaviour.
10. Verify motor rotation, current, noise, and vibration where applicable.
11. Use the stop control to stop the motor and confirm that it de-energises.
12. If an overload trip or abnormal condition occurs, isolate power before investigating.

3. Record test results, settings, and any defects according to site procedure.

⚠ Safety

- Installation, wiring, and maintenance must be performed by qualified electrical personnel.
- Never open or work on an energised starter unless an approved procedure specifically permits it.
- Use lockout/tagout and verify absence of voltage before touching conductors.
- Do not bypass overload relays, interlocks, protective earthing, or short-circuit protection.
- Keep hands and tools clear of exposed terminals and moving machinery.
- Follow electrical ratings and local regulations; a DOL starter can produce high motor starting current.

Care & storage

- Inspect contactor contacts, terminals, enclosure, and wiring at scheduled intervals.
- Check overload relay settings and function according to the maintenance plan.
- Keep the enclosure clean, dry, and free from dust buildup.
- Tighten terminals only to the specified torque with power isolated.
- Replace worn or damaged components with correctly rated approved parts.

Common mistakes to avoid

- Using a starter with an unsuitable motor or supply rating.
- Setting the overload relay incorrectly or bypassing protection.
- Working on terminals without isolation and voltage verification.
- Restoring power with loose wiring, missing covers, or tools inside the enclosure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.